

application you want to build. If you want something that helps with exact-match queries, like the names of all the people who were born in 1988, or a list of all authors who wrote books on a particular subject, then you're better off with databases, without any extra effort. However, if you want to answer queries that are less predictable and need proper analysis, like a list of documents in which a particular keyword occurs, or anything similar that is more flexible, Solr is there for you.

Before creating a database in a DBMS, you need to properly design a schema—and even minor changes in the structure of the data need to be reflected back into the database. This is not so with Solr—actually, it acts more like a NoSQL document store, where documents are in turn stored in the form of a Lucene index. Does that mean we can use it as a NoSQL database? Probably, but Solr hasn't been built for that purpose; it's more efficient as a search server, and we should let it remain that way.

Features and components

Carefully implemented instances of Solr can provide features that can match even those of Google—and what's more, it allows for further customisation, if needed; after all, it's open source! Let's take a brief look at some of the important and special features that Solr has to offer, without going into too much technical detail.

- **Faceted search:** This is one feature that stands out from the others. Whenever we encounter search in most applications, it comes with options to narrow down our search results with respect to certain parameters. Faceting provides such functionality, by providing facets as the navigational elements in the search. This technique is quite popular in places like online e-commerce websites, where we might need to filter search results from a particular category or a particular company.
- **Spatial search:** Although spatial search or geo-aware search is not a "feature" as such, it is quite relevant these days—and geo-location information is often included with the data, so it is important to mention it here. Spatial search is a plug-in that is expected to be included with native support in Solr version 4.0, and will provide results filtered according to the location information. For example, a search for a coffee shop should produce results suitable to your current location.
- **Clustering:** Clustering is also used for the aggregation of data. Solr provides for search result clustering with the Carrot2 clustering engine. Clustering is different from faceting, in the sense that it is done dynamically. We don't need to mark documents or specify fields to be categorised for clustering to take place; it is done automatically by recognising the words that occur most commonly in the text, and by recognising the structure of the text. Clustering is done when the search query is executed, while faceting is done during indexing. One of the clustering algorithms that I am aware of is the

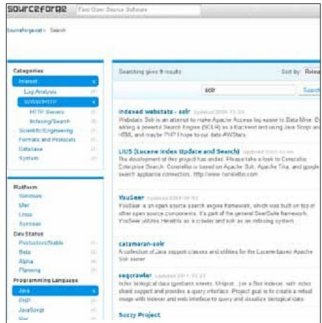


Figure 1: A live example of Facets in the search application of the SourceForge website

k-means clustering algorithm.

Other features (which search engines must obviously have) include *hit highlighting*, *auto-suggest*, *spell-checker* and the 'More like this' functionality, which do not need introduction.

Installation and set-up

First of all, let's get down to installing and setting up a single core instance of Solr on top of Tomcat. We will try to install the example schema that comes along with the archive. Before proceeding, it is worth mentioning that both the Lucene and Solr projects were merged in March 2010, due to which the version numbering sometimes becomes quite ambiguous. To avoid such confusion, let's look at the current and next version numbers of Lucene/Solr:

Version number	Status
1.4.1	The current version of Apache Solr
1.5 / 2.0	These were the next versions to be released before the Lucene/Solr merge, and have been skipped in order to keep up with Lucene.
3.1	This is the next Lucene/Solr point release.
4.0	This will be the next major release after the Lucene/Solr merge.

So, having dealt with that, you can begin by downloading and unzipping the *apache-solr-1.4.1.tgz* file from the website. As mentioned before, you will need to have at least Java 5 and a servlet container installed. In the *apache-solr-1.4.1/dist* directory, you will find the *apache-solr-1.4.1.war* file, which can be uploaded to any servlet